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Development of Fine-tuning Multimodal Large Language Models for Diabetic Retinopathy Analysis

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Introduction

Diabetic Retinopathy (DR) is a leading cause of blindness, affecting over 103 million people globally. In Thailand, only about 45.5% of the 3.6 million diabetic patients receive the necessary annual eye screenings, largely due to a shortage of ophthalmologists. While general-purpose Multimodal Large Language Models (MLLMs) like GPT-4o and Gemini show promise for image analysis, their reliability for clinical diagnosis is still uncertain. This research aims to first quantify the baseline performance of these general MLLMs and then develop a specialized, fine-tuned model using MED-Gemma to significantly improve diagnostic accuracy for DR, creating a reliable tool to support clinicians and expand screening accessibility.

Phase I : Baseline Evaluation of General MLLMs



This initial phase evaluated the out-of-the-box performance of leading MLLMs on DR diagnosis, based on the a quantitative evaluation study "Can Multimodal Large Language Models Diagnose Diabetic Retinopathy from Fundus images? ".

Methods

Ultra-widefield fundus images from patients seen at the University of California, San Diego data base, were graded for DR severity by 3 retina specialists using the ETDRS classification system to establish ground truth. Four MLLMs (ChatGPT-4o, Claude 3.5 Sonnet, Google Gemini 1.5 Pro, and Perplexity Llama 3.1 Sonar/Default) were tested using 4 distinct prompts. These assessed multiple-choice disease diagnosis, binary disease classification, and disease severity.

Results: Highly Variable & Unreliable

Cohort Ground Truth Disease Status & Severity (n=309 Eyes)

Diagnosis	Number of Eyes	Percentage
Overall Status		
Diabetic Retinopathy (DR)	217	70.2%
No DR	92	29.8%
Severity Breakdown		
Mild NPDR	89	28.8%
Moderate NPDR	83	26.9%
Severe NPDR	4	1.3%
Early PDR	22	7.1%
High-risk PDR	19	6.1%

Binary Classification Performance (DR vs. No DR)

Model	Accuracy	Sensitivity	Specificity
ChatGPT 4o	64.4%	53.9%	89.1%
Perplexity Llama	60.2%	48.8%	87.0%
Claude Sonnet 3.5	41.7%	17.9%	97.8%
Gemini 1.5 Pro	32.4%	4.2%	98.9%

DR Severity Grading Performance (Prompt 3.1)

Model	Accuracy	Sensitivity	Specificity
Perplexity Llama	41.1%	24.7%	86.4%
ChatGPT 4o	39.5%	22.9%	86.6%
Gemini 1.5 Pro	39.2%	22.4%	85.8%
Claude Sonnet 3.5	31.4%	18.4%	84.0%

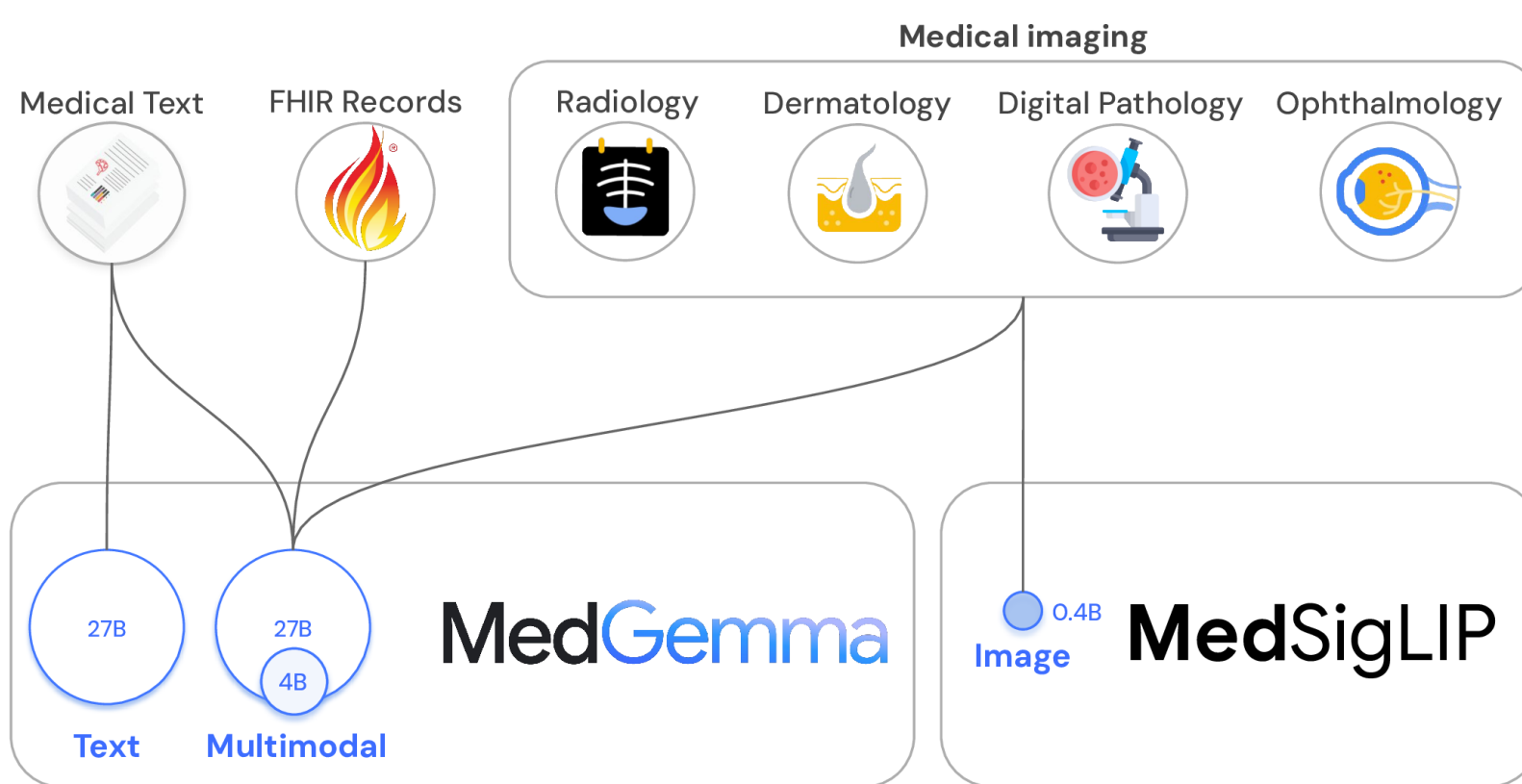
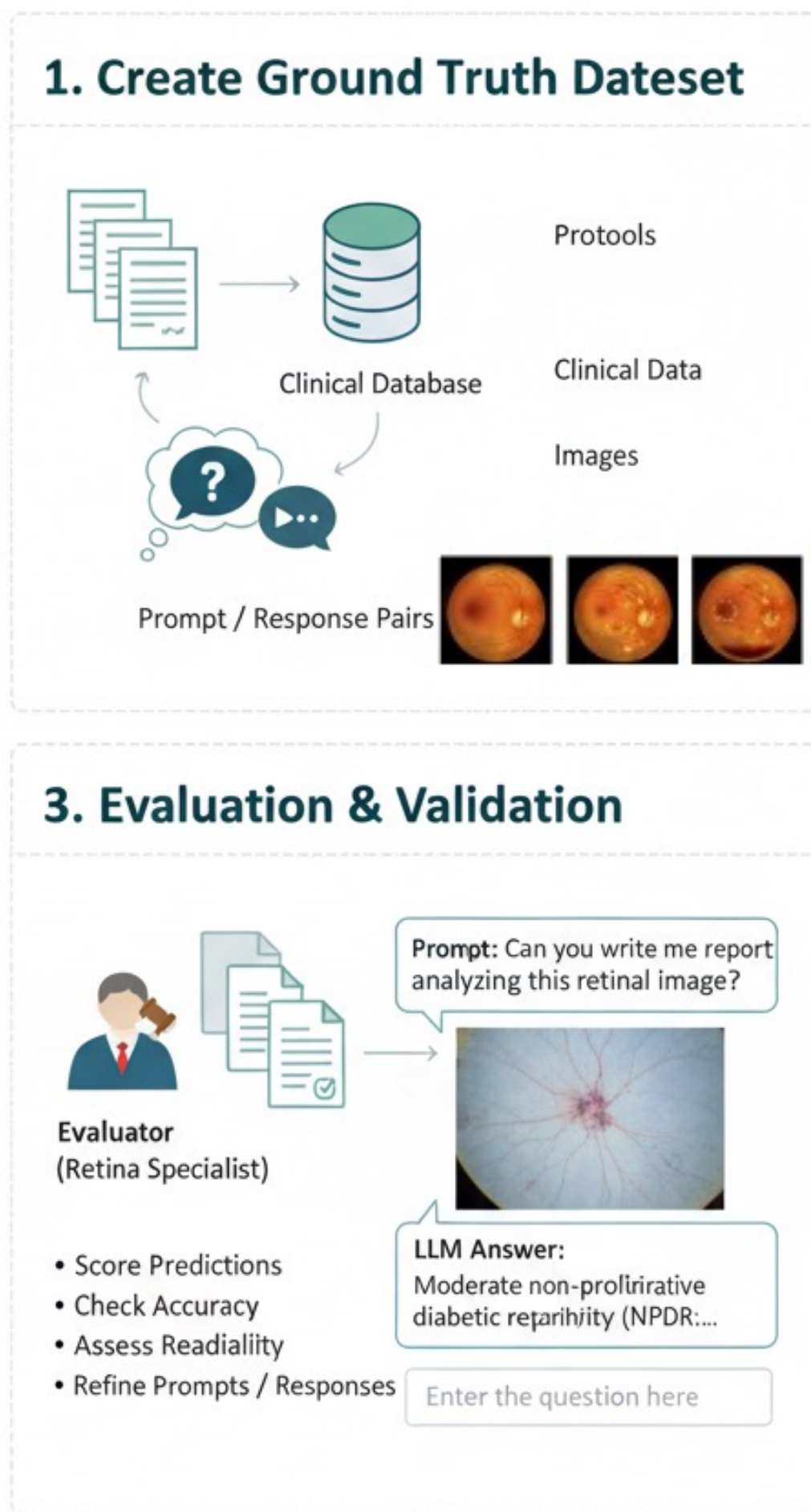
Conclusion (Phase 1)

Results Summary: From 309 eyes, 70.2% had DR. For binary classification, ChatGPT had the highest accuracy (64.4%), but sensitivity was highly variable and low across models (4.2% to 53.9%), while specificity was high (87.0%-98.9%). For severity grading, accuracy was poor (highest 41.1% by Perplexity) with very low sensitivity. While MLLMs show potential, their current diagnostic performance falls short of clinical standards for safe implementation in DR screening due to variability and low sensitivity.

Phase 2: Fine-Tuning with MED-Gemma model

To address the limitations identified in Phase 1, we fine-tuned Gemma, a state-of-the-art open model, using MED-Gemma, a version specialized for the medical domain.

Phase 2: Fine-Tuning Methodology



Results & Improvement

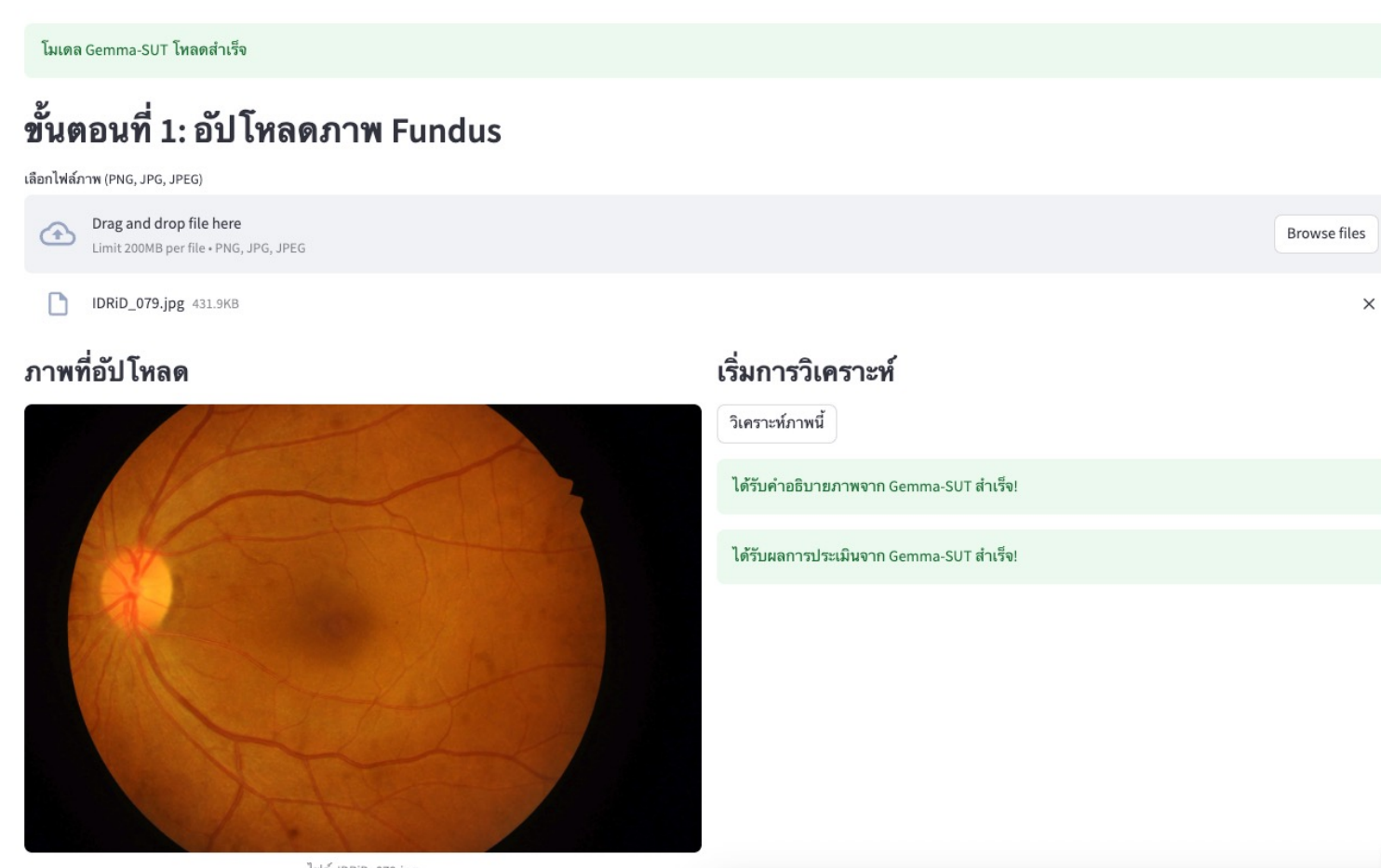


Application Prototype:

SUT¹ AIMS Artificial Intelligence & Modeling for Materials Science

เครื่องมือวิเคราะห์ภาพถ่ายจอประสาทตา (Fundus/OCT Image Analysis)

- คำสั่งและตัวอย่าง:
- เมื่อใช้งานโมเดล AI นี้ให้ระบุรายละเอียดให้ชัดเจนที่สุดเท่าที่
 - โมเดลจะวิเคราะห์และประมวลผลข้อมูลภาพทางการแพทย์ที่ส่งมาให้ได้
 - หากมีข้อผิดพลาดหรือข้อสงสัย กรุณาติดต่อทีมพัฒนา
 - ความแม่นยำของผลลัพธ์ขึ้นอยู่กับคุณภาพของภาพที่ส่งมาวิเคราะห์



ผลการประเมินจาก Gemma-SUT:

จากการวิเคราะห์ภาพถ่าย Fundus ที่ให้มา:

- ผู้ป่วยมีภาวะเบาหวานขึ้นตา (Diabetic Retinopathy) หรือไม่? ไม่พบหลักฐานของเบาหวานขึ้นตา
- ถ้าพบ ภาวะเบาหวานขึ้นตา (Diabetic Retinopathy) โปรดระบุระดับความรุนแรง: ปกติ

Overall Conclusion & Future Work

Specialized fine-tuning significantly improves MLLM performance, creating a path towards reliable AI-powered tools to assist in the fight against diabetic blindness.

Next Steps:

Conduct larger-scale clinical validation and Integrate into telemedicine platforms for remote deployment.